

Task 3: Install kubectl

```
allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
```

Task 4: Set Up Your Local Cluster

```
$ kind create cluster --name devops-cluster
Creating cluster "devops-cluster" ...
  • Ensuring node image (kindest/node:v1.35.0) ...
  ✓ Ensuring node image (kindest/node:v1.35.0)
  • Preparing nodes ...
  ✓ Preparing nodes
  • Writing configuration ...
  ✓ Writing configuration
  • Starting control-plane ...
  ✓ Starting control-plane
  • Installing CNI ...
  ✓ Installing CNI
  • Installing StorageClass ...
  ✓ Installing StorageClass
Set kubectl context to "kind-devops-cluster"
You can now use your cluster with:

kubectl cluster-info --context kind-devops-cluster

Not sure what to do next? Check out https://kind.sigs.k8s.io/docs/user/quick-start/

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$ docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
282fd2152fe   kindest/node:v1.35.0  "/usr/local/bin/entr..." About a minute ago    Up 54 seconds    127.0.0.1:49690->6443/tcp          devops-cluster-control-plane

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$
```

```
allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
devops-cluster-control-plane        Ready    control-plane   96s   v1.35.0

allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:49690
CoreDNS is running at https://127.0.0.1:49690/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

```
Non-terminated Pods: (9 in total)
Namespace           Name                                                    CPU Requests  CPU Limits  Memory Requests  Memory Limits  Age
-----
kube-system          coredns-7d764666f9-bsrzd                               100m (1%)    0 (0%)      70Mi (1%)       170Mi (4%)     3m46s
kube-system          coredns-7d764666f9-wnk4h                               100m (1%)    0 (0%)      70Mi (1%)       170Mi (4%)     3m46s
kube-system          etcd-devops-cluster-control-plane                       100m (1%)    0 (0%)      100Mi (2%)      0 (0%)         3m53s
kube-system          kindnet-x46x6                                           100m (1%)    100m (1%)   50Mi (1%)       50Mi (1%)      3m46s
kube-system          kube-apiserver-devops-cluster-control-plane              250m (3%)    0 (0%)      0 (0%)          0 (0%)         3m53s
kube-system          kube-controller-manager-devops-cluster-control-plane    200m (2%)    0 (0%)      0 (0%)          0 (0%)         3m54s
kube-system          kube-proxy-rbkhc                                         0 (0%)       0 (0%)      0 (0%)          0 (0%)         3m46s
kube-system          kube-scheduler-devops-cluster-control-plane              100m (1%)    0 (0%)      0 (0%)          0 (0%)         3m53s
local-path-storage   local-path-provisioner-67b8995b4b-h6xz5                0 (0%)       0 (0%)      0 (0%)          0 (0%)         3m46s
Allocated resources:
```

```
allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kubectl get namespaces
NAME                STATUS    AGE
default             Active    11m
kube-node-lease     Active    11m
kube-public         Active    11m
kube-system         Active    11m
local-path-storage  Active    11m
```

```
allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kubectl get pods -A
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	coredns-7d764666f9-bsrvd	1/1	Running	0	15m
kube-system	coredns-7d764666f9-wnk4h	1/1	Running	0	15m
kube-system	etcd-devops-cluster-control-plane	1/1	Running	0	15m
kube-system	kindnet-x46x6	1/1	Running	0	15m
kube-system	kube-apiserver-devops-cluster-control-plane	1/1	Running	0	15m
kube-system	kube-controller-manager-devops-cluster-control-plane	1/1	Running	0	15m
kube-system	kube-proxy-rbkhc	1/1	Running	0	15m
kube-system	kube-scheduler-devops-cluster-control-plane	1/1	Running	0	15m
local-path-storage	local-path-provisioner-67b8995b4b-h6xz5	1/1	Running	0	15m

```
allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kubectl get pods -n kube-system
```

NAME	READY	STATUS	RESTARTS	AGE
coredns-7d764666f9-bsrvd	1/1	Running	0	32m
coredns-7d764666f9-wnk4h	1/1	Running	0	32m
etcd-devops-cluster-control-plane	1/1	Running	0	32m
kindnet-x46x6	1/1	Running	0	32m
kube-apiserver-devops-cluster-control-plane	1/1	Running	0	32m
kube-controller-manager-devops-cluster-control-plane	1/1	Running	0	32m
kube-proxy-rbkhc	1/1	Running	0	32m
kube-scheduler-devops-cluster-control-plane	1/1	Running	0	32m

Task 6: Practice Cluster Lifecycle

```
allen@DESKTOP-CVA040U MINGW64 ~/k8s
$ kind delete cluster --name devops-cluster
Deleting cluster "devops-cluster" ...
Deleted nodes: ["devops-cluster-control-plane"]
```